STAT332: Assignment 4 Question 5

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This question is a follow up to Q4 of assignment #3, and is thus also concerned with making paper frogs. My daughter and I did a more thorough experiment, and investigated both the effects for dimensions (i.e. 10cm x 10cm vs. 15cm x 15cm vs. 20cm x 20cm) and colour (i.e. gray vs. orchid) on the jumping distance of paper frogs. After making the 6 paper frogs (3 dimensions x 2 colours) and practicing a few times, we individually launch each frog 5 times; for a grand total of 60 jumps.

```r
library(readxl)
frog_dat <- read_excel("data/frogs.xlsx")
frog_dat <- as.data.frame(frog_dat)
frog_dat$dimensions <- as.factor(frog_dat$dimensions)
frog_dat$colour <- as.factor(frog_dat$colour)
```

(i) produce the two-way ANOVA table, and perform the usual F tests. What are your conclusions on the effects of paper dimensions, colour and their interaction?

```r
fit <- aov(jump ~ dimensions * colour, data = frog_dat)
summary(fit)
```

```
##                   Df Sum Sq Mean Sq F value   Pr(>F)    
## dimensions         2  73601   36800  66.672 2.59e-15 ***
## colour             1   1030    1030   1.866    0.178    
## dimensions:colour  2     45      22   0.041    0.960    
## Residuals         54  29806     552                     
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Given that the p-value of paper dimension is much smaller than 0.05, we reject the null hypothesis. We conclude that there is very strong evidence that effect of paper dimension is highly significant in the jump distance.

The p-value for paper colour is greater than 0.05 so we fail to reject the null hypothesis. We conclude that there is strong evidence that the effect of paper colour does not have a statistically significant effect on jump distance.

The p-value for the interaction between dimension and colour is much greater than 0.05 so we fail to reject the null hypothesis. We conclude that there is very strong evidence that the effect of dimensions is consistent across both colours. More simply, colour does not change how size affects jumping.

(ii) Compute all relevant pairwise comparisons, using Tukey's method, and obtain their CI's with a family level of 95%.

```r
TukeyHSD(fit, "dimensions", conf.level = 0.95)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = jump ~ dimensions * colour, data = frog_dat)
##
## $dimensions
##                         diff        lwr        upr    p adj
## 15cmx15cm-10cmx10cm -61.605  -79.50977 -43.700226 0.000000
## 20cmx20cm-10cmx10cm -82.510 -100.41477 -64.605226 0.000000
## 20cmx20cm-15cmx15cm -20.905  -38.80977  -3.000226 0.018403
```

The CI for the comparison between (15x15 cm) and (10x10 cm) is [-79.51, -43.70]. This CI does not include 0 and has a p-value < 0.001 so we can say that 15x15 frogs jump significantly farther than 10x10 frogs.

The CI for (20x20 cm) and (10x10 cm) is [-100.41, -64.61]. This CI does not include 0 and has p-value $<$ 0.001 so we can sat that the 20x20 frogs jump significantly farther than 10x10 frogs.

Finally, the CI for the comparison between (20x20 cm) and (15x15 cm) is [-38.81, -3.00]. This CI does not include 0 and has p-value $0.018 < 0.05$ so we can say that there is statstical evidence that 20x20 frogs jump farther than 15x15 frogs.

(iii) Using part a.ii, determine which one (or ones) of groups jumps the furthest? From the comparisons, 20x20 $>$ 15x15 $>$ 10x10 with significant differences between each group. The 20x20 cm paper frogs jump the furthest on average, and all three sizes differ significantly from each other.

(b) Expand on the two-way ANOVA model fitted in part a by including person as a blocking factor. In other words, fit a two-way ANOVA model with dimensions and colour as factors, and person as a blocking factor.

(c) Produce the two-way ANOVA table, and perform the usual F tests. What are your conclusions on the effects of paper dimensions, colour, and their interaction? Are they the same as in a.i?

```r
block_fit <- aov(jump ~ dimensions * colour + person, data = frog_dat)
summary(block_fit)
```

```
##                   Df Sum Sq Mean Sq F value  Pr(>F)
## dimensions         2  73601   36800  76.863 < 2e-16 ***
## colour             1   1030    1030   2.151 0.14835
## person             1   4431    4431   9.254 0.00365 **
## dimensions:colour  2     45      22   0.047 0.95437
## Residuals         53  25375     479
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The p-value for dimensions is < 0.05 and is still highly significant just as in part a. However, we note that the F-value is increased from 66.7 to 76.9 which implies even stronger evidence after blocking.

The p-value for colour is > 0.05 so we conclude that there is no evidence that colour affects jump distance as in part a.

The p-value for person is < 0.05, so we reject the null hypothesis and conclude that there is strong evidence that the person launching the frog affects jump distance. This could explain the variation unexplained by the model in part a.

Lastly, the p-value for interaction between dimension and colour remains insignificant with p-value > 0.05. We thus fail to reject the null hypothesis and conclude that the interaction is highly insignificant.

(ii) In section 9.6 of the slides, I mentioned that estimating and comparing the blocks is usually not of interest. Though this is true, for this particular problem, my daughter nevertheless wants to know if she is better than I am at making the paper frogs jump further. Compute a 95% CI to test this.

Since person has 2 levels, the comparison is just the difference in mean jump distance between the daughter and you, adjusted for dimensions and colour.

```
fit2 <- lm(jump ~ dimensions * colour + person, data = frog_dat)
summary(fit2)
```

```
##
## Call:
## lm(formula = jump ~ dimensions * colour + person, data = frog_dat)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -54.427  -9.378   0.340   8.496  70.777
##
## Coefficients:
##                                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)                      109.437      7.474  14.643  < 2e-16 ***
## dimensions15cmx15cm              -60.230      9.785  -6.155 1.02e-07 ***
## dimensions20cmx20cm              -80.430      9.785  -8.219 5.00e-11 ***
## colourorchid                      10.590      9.785   1.082  0.28406
## personDaughter                    17.187      5.650   3.042  0.00365 **
## dimensions15cmx15cm:colourorchid  -2.750     13.839  -0.199  0.84324
## dimensions20cmx20cm:colourorchid  -4.160     13.839  -0.301  0.76489
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 21.88 on 53 degrees of freedom
## Multiple R-squared:  0.7571, Adjusted R-squared:  0.7296
## F-statistic: 27.54 on 6 and 53 DF,  p-value: 1.186e-14
```

```
confint(fit2, "personDaughter", level = 0.95)
```

```
##                   2.5 %   97.5 %
## personDaughter 5.854905 28.51843
```

Since the interval is positive and hence does not include 0, we conclude that the daughter's frogs jump significantly further than yours. Thus, the daughter is better at making the paper frogs jump farther.

(iii) Repeat a.ii and a.iii using the two-way ANOVA model with blocking. Do you reach the same conclusion about which one (or ones) of groups jumps the furthest?

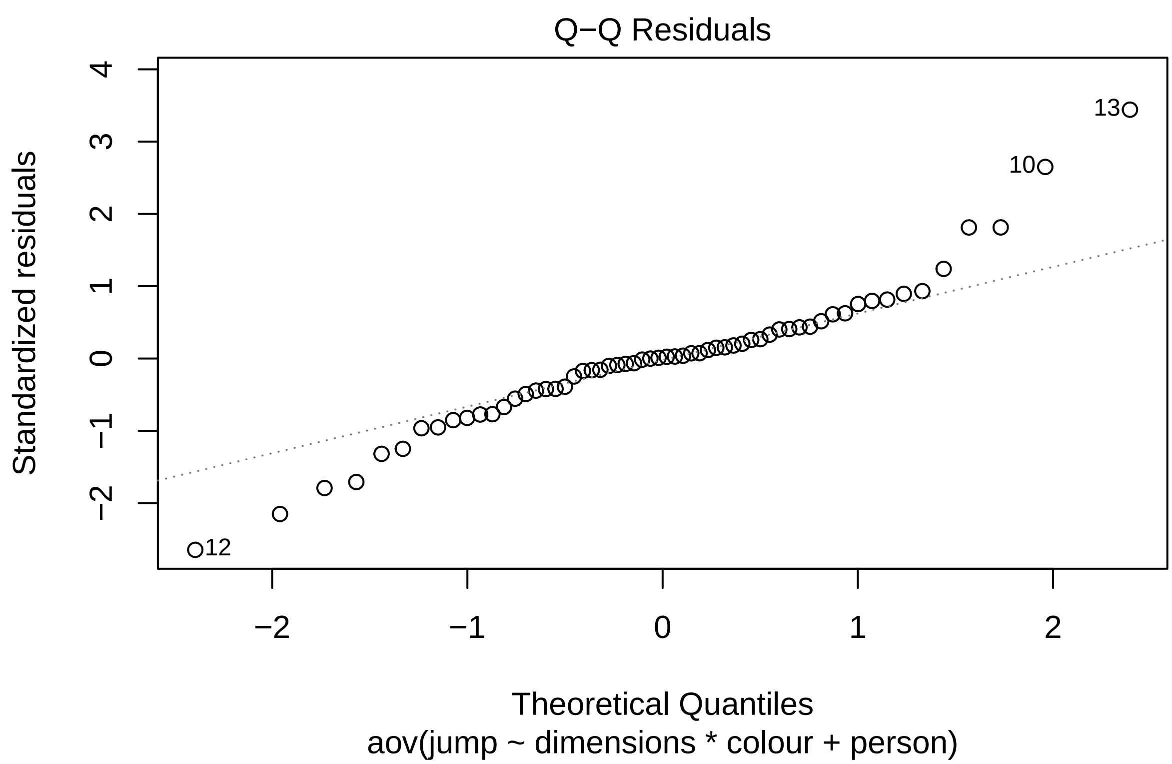
```
TukeyHSD(block_fit, "dimensions", conf.level = 0.95)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = jump ~ dimensions * colour + person, data = frog_dat)
##
## $dimensions
##                         diff       lwr        upr     p adj
## 15cmx15cm-10cmx10cm -61.605 -78.28945 -44.920547 0.0000000
## 20cmx20cm-10cmx10cm -82.510 -99.19445 -65.825547 0.0000000
## 20cmx20cm-15cmx15cm -20.905 -37.58945  -4.220547 0.0106336
```

Since the width of each interval changes by 1-2 cm, we can say that the same conclusion applies to the two-way ANOVA model with blocking. Therefore, we conclude that there is statistically significant differences between each group of paper dimensions such that (20x20) > (15x15) > (10x10) just as in part a.

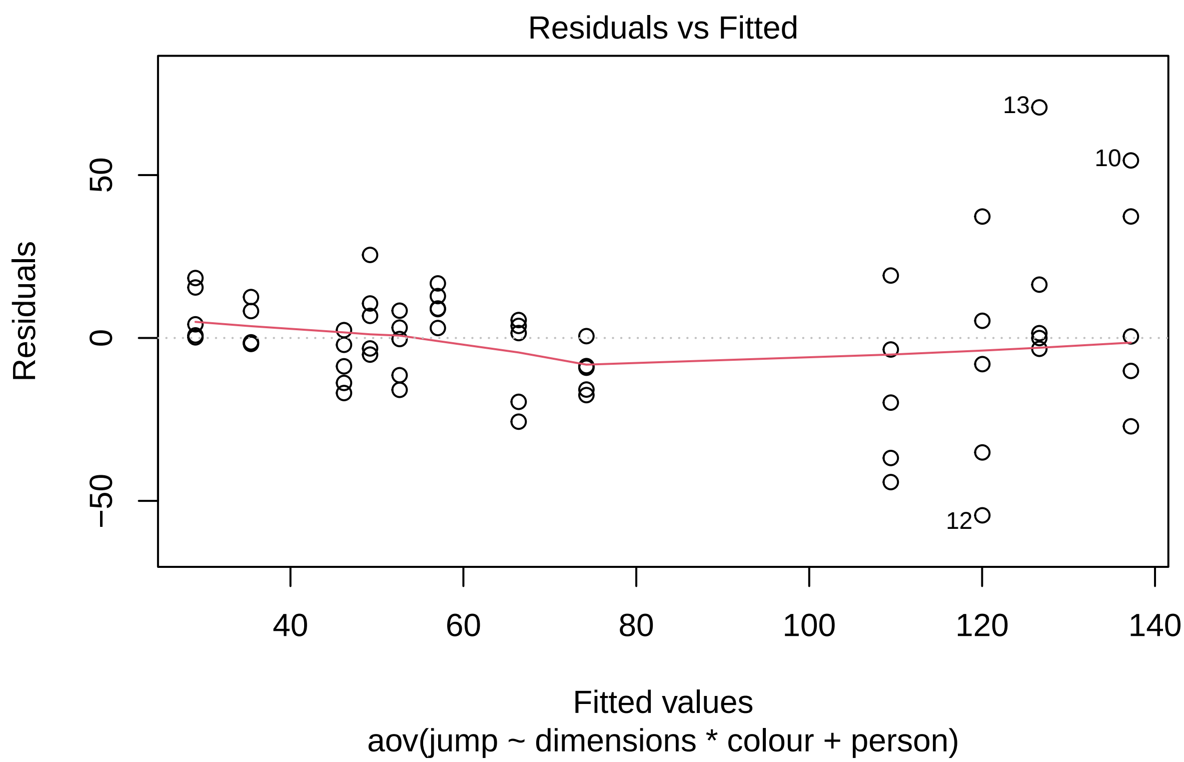
(iv) Check the assumptions of the ANOVA model, and look for possible outliers. IN particular, test the normality assumption, 0 expectation of errors and equality of variance.

```
# qq-plot of the residuals to check Normality assumption
plot(block_fit, which = 2)
```



This qq-plot shows that while the residuals are approximately normal in the center, there are noticeable deviations in the tails. This suggests mild departure from the normality assumption.

```r
# plot of residuals vs. fitted values to check zero expectation assumption as
# well as equality of variance assumption
plot(block_fit, which = 1)
```



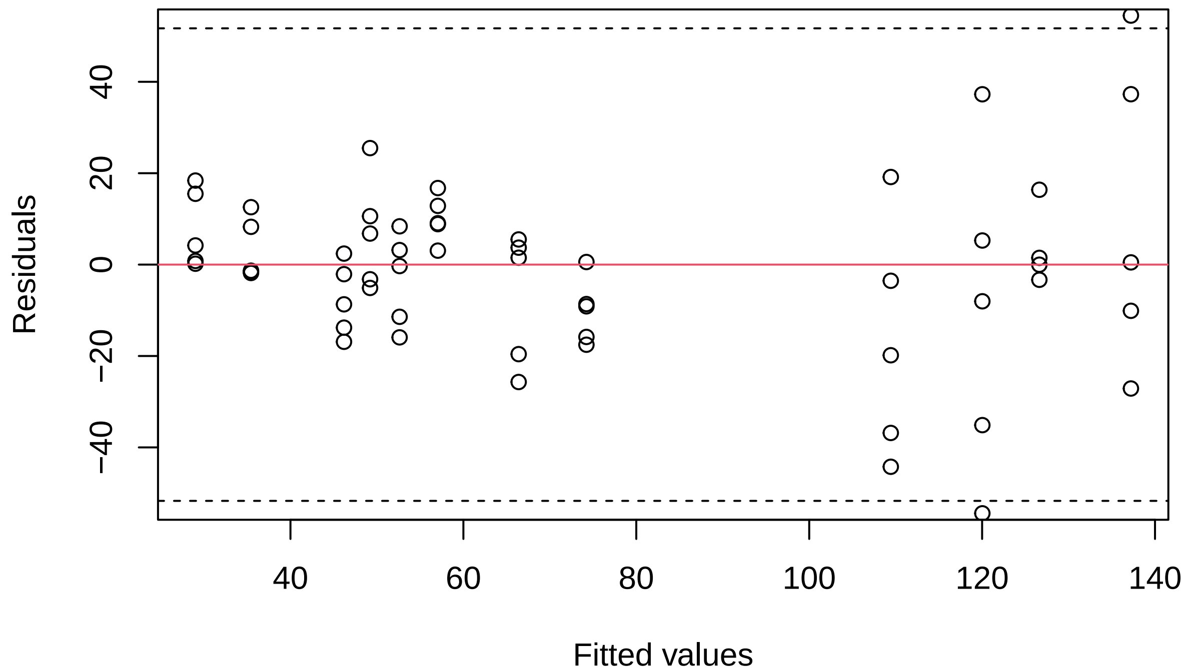
The residuals appear to be centered around zero, so the zero expectation assumption is satisfied. However, there is a slight funnel shape appearant which causes the residuals to increase with the fitted value. Thus, we cannot conclude that the constant variance assumption is satisfied.

```r
# check for the presence of outliers
frog.res <- data.frame(res = residuals(block_fit), fitted = fitted(block_fit))

mse <- sum(residuals(block_fit)^2) / block_fit$df

tmp <- qnorm(1 - 0.05/(2 * 60), sd = sqrt(mse * (1 - 1/2)))
plot(res ~ fitted, data = frog.res,
    xlab = "Fitted values",
    ylab = "Residuals",
    ylim = c(-tmp, tmp))

abline(h = 0, col = 2)
abline(h = tmp, lty = 2)
abline(h = -tmp, lty = 2)
```



Based on this plot, there is at least one potential outlier, as one residual falls outside the adjusted limits. However, most observations lie within the acceptable range, so the majority of the data appear consistent with the model.